

VSB ENGINEERING COLLEGE

Academic Performance & Parent Notification Platform
Comprehensive Step-by-Step User Manual (Admin & Staff)

1. Accessing the Academic Portal

The portal is accessible over both the live cloud deployment and the local workstation:

Access Endpoints

Live Cloud Portal: <https://remarkable-exploration-production-ab2a.up.railway.app>

Local Development: <http://localhost:5173> | API Health: </health> (Status: 200 HEALTHY)

The login screen features an interactive Role Switcher at the top:

- **Admin Login (Indigo Tab):** For Principal, Deans, HODs, and Office. Grants complete institutional access.
- **Staff Login (Emerald Tab):** For Subject Handlers and Mentors. Isolates mark entry strictly to assigned sections.

2. Master Demo Credentials

Role	Login Tab	Email / Username	Password	Default Portal
Administrator	Admin Tab	admin@college.edu	admin123	/admin/dashboard
Faculty (Email)	Staff Tab	sarah.cse@college.edu	Sarah@15081988	/staff/dashboard
Faculty (Username)	Staff Tab	sarah.jenkins	Sarah@15081988	/staff/dashboard

3. Administrator Walkthrough: Academic Setup

Before enrolling students or assigning faculty, the administrator must configure the core institution hierarchy:

Step 3.1: Configure Academic Hierarchy

- **Departments:** Navigate to "Academic Structure" -> "+ Add Department". Define Department Code (e.g., CSE, ECE, MECH) and full department title.
- **Academic Year:** Set the active working academic session (e.g., 2025-2026). Prior academic years can remain archived.
- **Years & Sections:** Create standard cohorts: Year 1 to Year 4, and link sections (Section A, B, C) per department.
- **Subjects Curriculum:** Click "+ Add Subject". Enter Subject Code (e.g., CS8501), Subject Name (e.g., Theory of Computation), Semester (1-8), and Maximum Marks (default 100.0).

Step 3.2: Add Faculty Members

- **Manual Faculty Entry:** Go to "Faculty Management" -> Click "+ Add Faculty". Enter Full Name, Official Email, Department, and Date of Birth (YYYY-MM-DD).
- **Password Standardization:** New faculty accounts automatically receive an institutional password using the format `FirstName@DDMMYYYY` (e.g. Sarah@15081988).
- **Teacher Assignment:** Map faculty to teaching loads: Select Staff Member -> Choose Department -> Select Year & Section -> Pick Subject -> Click "Assign".

4. Administrator Walkthrough: Student Management & Bulk Upload

Student records can be created individually or imported in batches using standard spreadsheet files:

Step 4.1: Bulk Excel / CSV Import (Cohort Upload)

- **Download Template:** Navigate to "Student Directory" -> "Bulk Import" -> Click "Download Sample CSV Template".
- **File Columns:** Ensure your Excel/CSV has the exact header columns: Register Number, Student Name, Department, Year, Section, Parent Name, Parent Mobile.
- **Duplicate Resolution:** Select how existing records should be treated: "Skip Duplicate" (keeps existing entries) or "Update Existing" (refreshes parent contact or section info).
- **Upload & Parse:** Drag and drop the spreadsheet file. The platform validates phone numbers and student register numbers, displaying a summary of parsed records.

Excel Upload Example Roster

921321104001, Aravind S, CSE, 3, A, Senthil Kumar, 9876543210
921321104002, Divya M, CSE, 3, A, Murugan P, 9876543211

Step 4.2: Individual Student Registration

- **Add Single Student:** Click "Student Directory" -> "+ Add Student". Fill in Register Number, Student Name, Department, Year, Section, and Parent Details -> Click "Save".

5. College Notices & Emergency Broadcasts

Publish circulars to parents and students across multiple channels:

- **Create Notice:** Navigate to "College Notices" -> Click "+ Create Notice".
- **Notice Type:** Select classification: Holiday Announcement, Exam Timetable, Semester Fee Circular, or Urgent Alert.
- **Target Audience:** Filter dispatch scope: Entire College, Specific Department, Specific Year, or Specific Section.
- **Dispatch Channels:** Choose delivery method: SMS, WhatsApp, or Both. The background queue dispatches notifications reliably.

6. Institutional Reports & PowerPoint Presentations

- **Official PDF Grade Sheets:** Navigate to "Reports & Analytics". Select Department, Year, and Assessment to export clean PDF rosters.
- **PowerPoint Slides (.pptx):** Click "Export to PPTX" to generate a presentation deck with class toppers and pass percentages for management meetings.

7. Staff / Faculty Portal Walkthrough

Faculty members access a role-restricted interface focused on their assigned teaching subjects:

Step 7.1: Staff Sign-In & Dashboard

- **Login Tab:** Click the Emerald "Staff Login" tab on the authentication screen.
- **Credentials:** Enter official staff email or username (e.g. sarah.cse@college.edu or sarah.jenkins) and password (Sarah@15081988).
- **Class Load Overview:** The dashboard displays assigned lecture classes (e.g. CSE - 3rd Year - Section A | CS8501).

Step 7.2: Mark Entry

- **Select Class:** Click "Enter Marks" on the corresponding class card.
- **Select Assessment:** Choose assessment: Internal Assessment 1 (IA-1), IA-2, IA-3, Model Exam, or Semester Exam.
- **Enter Marks:** The student list loads automatically. Enter marks scored out of maximum marks (e.g., 85/100).
- **Real-Time Pass/Fail:** The system dynamically evaluates grades and marks scores below 50.0 as Critical/Fail in red badges.
- **Persist Marks:** Click "Save Marks" to record scores directly in the PostgreSQL database.

Step 7.3: Mark Corrections & Audit Trail Compliance

To meet NBA and NAAC academic compliance standards, any modification to a previously entered mark triggers audit logging:

- **Modification Modal:** When editing an existing mark, the system prompts for a mandatory "Reason for Modification".
- **Audit Logging:** The previous score, revised score, staff identity, timestamp, and reason are recorded permanently.
- **Audit Review:** Administrators can inspect all grade alterations in the "Audit Logs" viewer.

Step 7.4: Sending Automated Parent Notifications

- **Review Scores:** Once marks are finalized, click "Notify Parents" on the mark sheet.
- **Preview Message:** A live preview shows the exact SMS/WhatsApp text that will be sent to the parent.
- **Dispatch:** Click "Send to All Parents". Parents receive the mark alert within seconds.

Ø=Üñ Sample Automated Parent Message

Dear Parent, your ward Divya M has scored 88/100 in CS8501 (Internal Assessment 1). Overall Status: PASS.
- Examination Office, VSB Engineering College